Standardized Benchmarking of Multi-Agent Distributed Machine Learning in Augmented Reality

Research Update #2

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1. Progress Summary

Our core research question is: Can a multi-agent, edge-distributed AR pipeline deliver accurate, context-aware educational overlays in real time on commodity mobile hardware? This update directly addresses the feedback received on Update #1. We have added feasibility details and download sources for all datasets, so they can be worked with soon. Each paper cited now notes whether it includes code, models, or datasets (e.g., MultiAgentBench has a public GitHub repo, SAM 2 and YOLOv8 both have Apache 2.0 repos on GitHub and HuggingFace). All references are now listed in full at the end of this document. We have also replaced GPT-4o-mini with Phi-3-mini (on-device) and GPT-4.1-mini (cloud fallback), given its potential deprecation. Finally, a 4-week timeline with per-student delegation has been added. The AR Flashcard Tutor pipeline is ready to move into implementation.

2. Datasets

All datasets below have been verified as publicly available and feasibly downloadable.

Dataset	Purpose	Availability & Size
ImageNet (ILSVRC)	Object recognition pre-training	image-net.org (academic registration required). ILSVRC subset ~50 GB
COCO	Object detection & segmentation	cocodataset.org. ~20 GB, freely available. GitHub repo and active leaderboard.
Open Images V7	Extended detection with visual relationships	github.com/openimages. Subsets downloadable via AWS or TensorFlow Datasets (~500 GB full; subset feasible).
SQuAD 2.0 / TriviaQA	Quiz generation fine-tuning	HuggingFace: rajpurkar/squad_v2 and mandarjoshi/trivia_qa. Small files, easily downloaded.
Custom Flashcard Dataset	Domain-specific card recognition	Self-collected 500–1,000 flashcards. To be photographed and annotated by the team in Week 2.

3. Models(Updated)

Component	Model	Code / Data Availability
Object Detection	YOLOv8-Nano	GitHub: github.com/ultralytics/ultralytics (Apache 2.0). Weights on HuggingFace: ultralytics/assets.
OCR	PaddleOCR (on-device)	GitHub: github.com/PaddlePaddle/PaddleOCR (Apache 2.0). Supports mobile deployment.
Explanation + Quiz Gen	Phi-3-mini (on-device) / GPT-4.1-mini (cloud)	Phi-3-mini on HuggingFace: microsoft/Phi-3-mini-4k-instruct. GPT-4.1-mini via OpenAI API.
Visual Annotation	SAM 2 (Segment Anything Model 2)	GitHub: github.com/facebookresearch/segment-anything-2 (Apache 2.0). HuggingFace: facebook/sam2-hiera-tiny.

4. Timeline & Delegation

Week	Alexis	Parthkumar	Joint Milestone
Week 1	Will download the YOLOv8-Nano repository from Ultralytics GitHub and verify installation. Will run the pre-trained COCO weights on 10 sample images to confirm the environment works.	Will download the COCO validation split (~5,000 images) and SQuAD 2.0 from HuggingFace. Will install PaddleOCR and test it on 5 printed flashcard photos to confirm text extraction works.	Both environments will be set up and verified. Each student will be able to run their respective tool (YOLO / PaddleOCR) independently and produce output.
Week 2	Will run YOLOv8-Nano on a batch of 50 COCO validation images. Will record detection accuracy (mAP) and inference time per image, and document which object categories the model handles well vs. poorly.	Will run PaddleOCR on 20 photographed flashcards (mix of typed and handwritten). Will record character-level accuracy and note failure cases such as blurry text, colored backgrounds, and small fonts.	Baseline results will be compiled into a shared spreadsheet. Both students will analyze where each model succeeds and fails, and identify which flashcard styles and object types will work best for the demo.

Week 3	Will photograph 250 flashcards across 2 subjects (biology, chemistry) under varying lighting and angles. Will upload to Roboflow and annotate with bounding boxes and class labels.	Will photograph 250 flashcards across 2 subjects (history, general science) and annotate in Roboflow. Will also test how YOLOv8-Nano performs on raw flashcard photos without any fine-tuning to establish a zero-shot baseline.	A 500-image custom flashcard dataset will be complete and annotated. Zero-shot detection results will be documented to provide a before vs. after comparison once fine-tuning begins.
Week 4	Will fine-tune YOLOv8-Nano on the custom flashcard dataset using Google Colab. Will experiment with training for 25, 50, and 100 epochs and record mAP at each checkpoint.	Will compare the fine-tuned model's detection accuracy against the Week 2 zero-shot baseline. Will run the fine-tuned model on a held-out test set of 50 flashcards and document per-class accuracy.	A fine-tuned YOLOv8-Nano model will be ready. A side-by-side comparison table showing pre-trained vs. fine-tuned accuracy on custom flashcards will be produced.

5. Research Paper Discussions

The following section discusses four key papers that we are using to layout the ground work for our project. Each paper is evaluated on its strengths, the problem it identifies, its inputs/outputs/metrics, its weaknesses and shortcomings, and how it directly connects to our AR Flashcard Tutor pipeline. We are using the guidelines the professor provided us for our homework assignments.

5.1 MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents (Zhu et al., 2025)

Strength of the Work: MultiAgentBench is one of the first benchmarks to evaluate LLM-based multi-agent systems across both collaborative and competitive dynamics simultaneously. Its MARBLE framework supports multiple coordination topologies — star, chain, tree, and graph — giving researchers a flexible, modular way to test how agent communication structure affects task performance.

Identifying the Problem: Existing single-agent benchmarks such as AgentBench, GAIA, and HumanEval do not capture the dynamics that emerge when multiple LLM agents must coordinate, divide labor, and sometimes compete. Multi-agent systems introduce challenges around communication overhead, emergent behavior, and scalability that single-agent evaluations completely ignore.

Inputs / Outputs / Metrics: Inputs include task descriptions, agent personas, domain databases, and communication topology configurations. Outputs are task completions evaluated against ground truth trajectories. Metrics include the milestone-based KPI (measuring partial task progress), collaboration scores (structured planning and communication quality), competition scores for adversarial scenarios, and final task completion accuracy. The paper also reports ablations across coordination protocols to show how topology choice affects performance.

Weaknesses & Shortcomings: The benchmark scenarios, while diverse, are largely synthetic or LLM-generated with human verification, which may not fully reflect the complexity of real-world deployment. The cognitive planning improvement of 3% in milestone achievement is decent, but it raises questions about whether the gains are meaningful. Additionally, the paper does not deeply explore how benchmark results degrade as agent count increases, leaving scalability largely untested.

Connection to our project: Since our project relies on a multi-agent edge-distributed architecture in which different agents (detection, OCR, explanation generation) must coordinate. This paper informs how we should think about measuring our own pipeline's agent interactions.

5.2 GAIA: A Benchmark for General AI Assistants (Mialon et al., 2023)

Strength of the Work: This paper has a different approach for benchmarking AI assistants: rather than targeting tasks that are professionally difficult for humans, it focuses on tasks that are conceptually simple for humans but require accurate, multi-step execution from AI systems. This made it so that the data is easy to evaluate and less prone to errors. Since all answers are factual, concise, and unambiguous. The three-level difficulty structure (Level 1–3) creates a useful gradient for measuring how well AI systems handle increasing reasoning chain complexity.

Identifying the Problem: Existing AI benchmarks such as MMLU and GSM8k are becoming saturated due to rapid LLM improvement and potential data leakage, making them poor indicators of true general capability. Model-based evaluation (using a stronger LLM as judge) is self-referential and introduces subtle biases. The paper targets the gap between impressive performance on narrow academic tasks and the broader, tool-augmented, real-world reasoning that a useful general AI assistant would need to perform.

Inputs / Outputs / Metrics: Inputs are real-world questions that may include images, web browsing requirements, multi-step reasoning, and tool use. Outputs are short, factual, unambiguous answers. The primary metric is exact-match accuracy, evaluated at each of the three difficulty levels. The benchmark demonstrates a stark disparity: human respondents average 92% accuracy while GPT-4 with plugins achieves around 15%, highlighting a large remaining gap between current AI capability and human-level general task solving.

Weaknesses & Shortcomings: GAIA's 466-question dataset is relatively small, the benchmark's limited scope may not be representative of the full range of real-world assistant use cases. The reliance on factual, unambiguous answers by design excludes open-ended reasoning or generative tasks that are increasingly important in deployed AI systems. Additionally, questions tied to web browsing may become stale as the web changes over time.

Connection to Our Project: This paper really covers the weakness on the previous paper discussed on section 5.1, the way we can implement the teaching in this paper is that our pipeline must not just detect an object but chain detection → OCR → explanation generation → AR overlay in a coherent way. The idea of measuring partial task completion also motivates our choice to track per-stage latency and accuracy separately rather than only measuring end-to-end output quality.

5.3 The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of LLMs (Patil et al., 2025)

Strength of the Work: This research provides a comprehensive benchmark for evaluating LLM function-calling, covering 5,551 question-function-answer pairs across Python, Java, JavaScript, REST APIs, and SQL. One of the main talking points was the Abstract Syntax Tree (AST) substring matching, which enables scalable, deterministic evaluation without requiring live function execution, a significant practical advantage for large-scale benchmarking.

Identifying the Problem: Existing function-calling benchmarks either focus exclusively on single-turn interactions, rely on narrow domain coverage that is prone to overfitting, or use LLM-based judges that introduce self-referential bias. BFCL identifies that state-of-the-art models perform well on isolated function calls but struggle with memory management and stateful multi-turn reasoning.

Inputs / Outputs / Metrics: Inputs are natural language user queries paired with structured function documentation. Outputs are function call predictions, including function name and parameter values. Evaluation metrics include AST-based accuracy, execution response matching, state-based and response-based checking for multi-turn scenarios, and exact-match accuracy for agentic tasks.

Weaknesses & Shortcomings: Memory management category reveals that even the top-performing model (o1-2024-12-17) achieves only 12% accuracy, suggesting the benchmark may currently be too difficult to drive meaningful model differentiation in that category. The AST evaluation, while scalable, requires restricting model outputs to Python-callable format, which may penalize models that produce correct answers in non-standard syntax.

Connection to Our Project: This research paper is closely related to our project, it provides an effective way on how we can approach a benchmarking method for our project. BFCL's multi-turn evaluation framework, specifically its state-based and response-based checking, is a direct model for how we will validate that our agents correctly chain function calls across pipeline stages.

5.4 Multi-Modal Multi-Task Federated Foundation Models for XR (Yan et al., 2025)

Strength of the Work: This paper systematically articulates the intersection of federated learning, multi-modal foundation models, and extended reality (XR) systems. Its SHIFT framework (Sensor diversity, Hardware heterogeneity, Interactivity, Functional variability, Temporality) provides a principled taxonomy of the challenges that make deploying large models on XR hardware fundamentally different from standard server-side deployment. The paper also proposes concrete evaluation metrics and dataset requirements specific to resource-constrained XR environments, filling a gap that generic ML benchmarking literature ignores.

Identifying the Problem: XR applications like AR collect deeply personal, multi-modal data (spatial layouts, gestures, gaze) that cannot be sent to centralized cloud servers due to privacy constraints and latency requirements. The paper frames this as a federated learning problem where edge devices must collaboratively train multi-modal, multi-task foundation models without centralizing sensitive data.

Inputs / Outputs / Metrics: unlike the other papers, this is not an empirical study so it does not report a single experimental result. Instead, inputs are characterized as multi-modal XR streams (RGB, depth, IMU, gaze, audio) and outputs are task predictions (object detection, segmentation, language generation). Proposed evaluation metrics include task-specific accuracy per modality, communication overhead, inference latency, privacy leakage bounds, and personalization quality, metrics that collectively capture the resource-accuracy tradeoff specific to edge XR deployment.

Weaknesses & Shortcomings: the work lacks experimental validation since it is not an empirical study. The SHIFT framework, while useful conceptually, does not yet come with a benchmark dataset or evaluation protocol, making it difficult to measure progress against the challenges it defines.

Connection to our project: The paper uses a mobile AR device running multiple models (YOLOv8-Nano, PaddleOCR, Phi-3-mini) under hardware constraints. This is very closely related to what we are aiming to do for our project. The SHIFT dimensions map directly onto challenges we face: hardware heterogeneity across iOS and Android devices, functional variability between detection and language tasks, and temporal variability as flashcard content changes across sessions.

Repo: https://github.com/parthj100/AR_Flashcards

6. References

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